

EUIJIN BAEK

Ph.D. Student, Computing Science — University of Alberta

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EDUCATION

Ph.D. in Computing Science

University of Alberta

📅 Sep 2024 – Present 📍 Edmonton, Canada

- Advisors: Prof. Mi-Young Kim, Prof. Randy Goebel
- Coursework: Reinforcement Learning I; Machine Learning Essentials
- GPA: 3.9/4.0

M.Sc. in Statistics

University of Toronto

📅 Sep 2023 – Jun 2024 📍 Ontario, Canada

- Coursework: Online Learning and Sequential Decision Theory; Statistical Methods for Machine Learning
- GPA: 3.97/4.0

Additional Coursework

Athabasca University

📅 Aug 2022 – Mar 2023 📍 Distance Education

- Linear Algebra I; Design and Analysis of Algorithms; Operating Systems
- GPA: 4.0/4.0

B.L. in Law & B.Sc. in Statistics

Dongguk University

📅 Mar 2013 – Aug 2021 📍 Seoul, South Korea

- Mathematical Statistics I, II; Regression Analysis; Exploratory Data Analysis; Mathematics for Statistics and R Exercises; Python Programming Basics
- GPA: 3.85/4.5

WORK EXPERIENCE

Data Scientist & ML Engineer

EineBlume Co. Ltd.

📅 Aug 2022 – Apr 2023 📍 Seoul, South Korea

- Develop and execute algorithms to optimize user engagement on Web Fiction Platform.

Data Scientist & ML Engineer

DataMKT Korea Inc.

📅 Mar 2021 – Aug 2022 📍 Seoul, South Korea

- Plan & develop algorithms for constructing Digital Marketing Platform.

PUBLICATIONS

TRM-Planner: Offline Target Planning and Distillation for Tiny Recursive Models

ACL 2026

Authors: **Euijin Baek**, Housam Babiker, Mi-Young Kim, Randy Goebel.

openreview.net/forum?id=ggkeH4cfjE

Corruption-Robust Variance-aware Algorithms for Generalized Linear Bandits under Heavy-tailed Rewards

UAI 2025 (PMLR 286: 4826–4843)

Authors: Q. Yu, E. Baek, X. Li, Q. Sun.

proceedings.mlr.press/v286/yu25b.html

PROJECT EXPERIENCE

** Click each project title to open details.*

[HealthLink]

University of Alberta

📍 Edmonton, Canada

Working on research project enhancing reasoning ability of Large Language Models.

[Online Learning] Corruption-Robust Variance-aware Algorithms for Generalized Linear Bandits under Heavy-tailed Rewards

University of Toronto

📅 Oct 2023 – Sep 2024

📍 Ontario, Canada

- Investigate heavy-tailed adversarial reward with corruption terms (e.g., click fraud in PPC).
 - Derive regret bound under adversarial corruption with known reward variance.
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[Recommender System] Web Fiction Recommender System

EineBlume Co. Ltd.

📅 Aug 2022 – Dec 2022

📍 Seoul, South Korea

- GRU-based recommendation using image, text, and time-dependent user interest.
 - Deployed training pipeline on SageMaker; data pipeline with Airflow and Redis (Feature Store).
 - ~2M interactions; +4.30% content consumption; +4pp retention; Average Precision@25 = 40%.
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[NLP - GPT] (Personal Project) Chatbot with GPT

Personal

📅 Nov 2022 – Dec 2022

- Developed chatbot powered by GPT; 170k dialogue-formatted web fiction samples as training data.
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[Reinforcement Learning] Social Media Content Recommender System

DataMKT Korea Inc.

📅 Mar 2022 – May 2022

📍 Seoul, South Korea

- Actor-Critic recommender for Instagram, Facebook, YouTube.
 - Implemented “Top-K Off-Policy Correction for a REINFORCE Recommender System” in PyTorch.
 - Kafka for streaming user interaction data.
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[NLP - NER] Extracting “New” company names (w. Samsung Electronics)

DataMKT Korea Inc.

📅 Oct 2021 – Dec 2021

📍 Seoul, South Korea

- BERT-based NER to extract new company names; reporting for Samsung Electronics.
 - Gold medal in Samsung competition; plaque awarded; contract total \$483,637.
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[NLP - Classification] Open-world Semi-Supervised Learning of Garbage Documents Classification

DataMKT Korea Inc.

📅 Nov 2021 – Dec 2021

📍 Seoul, South Korea

- Semi-supervised BERT classification for review/stock/real estate/advertisement.
- 9k labeled + 500k unlabeled; 90% accuracy; 450k new daily docs classified; marketer reports.

Social media channel & content category classification using API and transformer

DataMKTKorea Inc.

📅 Apr 2021 – Sep 2021

📍 Seoul, South Korea

- LDA-based channel/category setup; supervised kcelectra classification; 92% accuracy.

Brand/General SNS channel using n-gram

DataMKTKorea Inc.

📅 Mar 2021 – Apr 2021

📍 Seoul, South Korea

- n-gram sequence-matching classifier to distinguish brand vs general channels; 84% accuracy.

[Prediction] Hot Influencer After 30 days

DataMKTKorea Inc.

📅 Jun 2021 – Jul 2021

📍 Seoul, South Korea

- LSTM forecast of channel fan counts using 7,200 channels & 30-day engagement; 90% precision (tolerance-based).

(Personal Project) Real estate policy & house price with multiple linear regression

Personal

📅 Nov 2020 – Dec 2020

- PCA and regression to estimate policy impact on house prices (macro indices + policy features).

(Personal Project) Automatic stock & crypto trading system

Personal

📅 Jun 2022 – Apr 2023

- Time-series data collection via API; ML/RL price prediction; SageMaker deployment.
- Automatic crypto trading system based on predictions; earned \$40 per month.

[Explain AI] (Personal Project) Automatic Keyword Extraction to build keyword dictionary

Personal

📅 Jul 2021 – Oct 2021

- Applied LIME to prior classifier to extract per-category keywords; auto-updated dictionary.

SKILLS

Python R SQL TensorFlow Keras PyTorch NLTK Mecab Transformer spaCy Django REST
Docker Kafka Airflow on Kubernetes SageMaker Linux AWS IELTS 8.0